Year 10 Mathematics

Algebra 1

2.2 Simplifying Expressions: Addition and Subtraction

Simplifying Expressions: Addition and Subtraction

STARTER · 5 MIN

Date: _____ **Title:** Simplifying Expressions: Addition and Subtraction

LEARNING OBJECTIVES

- Identify like terms
- Simplify expressions by collecting like terms
- Simplify expressions with powers and more than one letter

STARTER *Answer in your book. No calculators.*

1. Work out $7 - 10$
2. Work out $-4 + (-6)$
3. Work out $-5 - (-8)$
4. A rectangle is 6 cm by 4 cm. Find its perimeter.

Starter — Answers

STARTER

Tick or correct in green pen

1. $7 - 10 =$ **-3**

2. $-4 + (-6) =$ **-10**

3. $-5 - (-8) = -5 + 8 =$ **3**

4. $P = 6 + 4 + 6 + 4 =$ **20 cm**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can identify like terms	☐
SILVER	I can simplify expressions by collecting like terms	☐
GOLD	I can simplify expressions with powers and more than one letter	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Collecting Like Terms

INTRODUCTION · 3 MIN

Like terms have exactly the same letters and the same powers.

Term

Numbers and letters multiplied together.

$5x$, $3ab$, $-2y^2$

Like terms

Same letters and the same powers.

$3x$ and $7x$ ✓ $3x$ and $3x^2$ ×

Collect

Add or subtract the numbers in front.

$4a + 2a - a = 5a$

Signs

The sign in front belongs to the term.

$5x - 2y - 3x = 2x - 2y$

Examples 1 and 2 — Collecting Like Terms

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Simplify

$4a + 3b + 2a - b$

Example 2

Simplify

$5x - 2y - 8x + 7y$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

a terms: $4a + 2a = 6a$

b terms: $3b - b = 2b$

Answer: $6a + 2b$

Tip: b means 1b.

Example 2

x terms: $5x - 8x = -3x$

y terms: $-2y + 7y = 5y$

Answer: $-3x + 5y$

Tip: the sign in front of a term moves with it.

Task 1

PRACTICE · 9 MIN

BRONZE *Simplify each expression*

1. $3x + 5x$

2. $9y - 4y + y$

3. $2a + 3b + 4a + b$

4. $7p - 2q - 3p + 5q$

5. A triangle has sides $2x$, $3x + 1$ and $x + 4$. Write an expression for its perimeter. Simplify your answer.

★ Simplify $x + x + x + y + y$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $3x + 5x =$ **$8x$**

2 $9y - 4y + y =$ **$6y$**

3 $2a + 3b + 4a + b =$ **$6a + 4b$**

4 $7p - 2q - 3p + 5q =$ **$4p + 3q$**

5 $P = 2x + 3x + 1 + x + 4 =$ **$6x + 5$**

★ $x + x + x + y + y =$ **$3x + 2y$**

Examples 3 and 4 — Powers and More Letters

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

Simplify

$3x^2 + 4x - x^2 + 2x$

Example 4

Simplify

$5ab + 2a - 3ba + 4a$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

x^2 terms: $3x^2 - x^2 = 2x^2$

x terms: $4x + 2x = 6x$

Answer: $2x^2 + 6x$

Tip: x^2 and x are not like terms.

Example 4

ab terms: $5ab - 3ab = 2ab$

a terms: $2a + 4a = 6a$

Answer: $2ab + 6a$

Tip: ab and ba are the same term.

Task 2

PRACTICE · 9 MIN

SILVER

Simplify each expression

1. $4x^2 + 3x^2$
2. $6y^2 - 2y + y^2 + 5y$
3. $3xy + 2yx - xy$
4. $2a^2 + 3a - 5 - a^2 + a + 8$
5. $4m^2n - mn^2 + 2nm^2 + 3mn^2$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $4x^2 + 3x^2 =$ **$7x^2$**

2. $6y^2 - 2y + y^2 + 5y =$ **$7y^2 + 3y$**

3. $3xy + 2yx - xy =$ **$4xy$**

4. $2a^2 + 3a - 5 - a^2 + a + 8 =$ **$a^2 + 4a + 3$**

5. $4m^2n - mn^2 + 2nm^2 + 3mn^2 =$ **$6m^2n + 2mn^2$**

Examples 5 and 6 — Exam-Style Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Example 5 (2 marks)

A rectangle has length $3x + 2$ and width $x - 1$.

Write an expression for its perimeter. Simplify your answer.

Example 6 (2 marks)

Simplify fully

$5p^2q - 3pq^2 + 2pq - p^2q + 4pq^2$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$3x + 2 + x - 1 + 3x + 2 + x - 1$

x terms: $8x$ numbers: 2

Perimeter **$= 8x + 2$**

Tip: a rectangle has two lengths and two widths.

Example 6

p^2q: $5p^2q - p^2q = 4p^2q$

pq^2: $-3pq^2 + 4pq^2 = pq^2$

Answer: $4p^2q + pq^2 + 2pq$

Tip: p^2q and pq^2 are different terms.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

1. Simplify $7c - 3d + 2c + 5d$ (2 marks)

2. Simplify $2x^2 - 5x + 3 - 4x^2 + x - 7$ (2 marks)

3. Simplify $3ab - 2a + 5ba - a$ (2 marks)

4. A triangle has sides $2y + 3$, $4y - 1$ and $y + 5$. Show that its perimeter is $7y + 7$ (2 marks)

5. Find the missing term: $5x + 2y + \square = 8x - y$ (2 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $7c - 3d + 2c + 5d =$ **$9c + 2d$**

2. $2x^2 - 5x + 3 - 4x^2 + x - 7 =$ **$-2x^2 - 4x - 4$**

3. $3ab - 2a + 5ba - a =$ **$8ab - 3a$**

4. $P = 2y + 3 + 4y - 1 + y + 5 =$ **$7y + 7$**

5. Missing term $= 8x - y - 5x - 2y =$ **$3x - 3y$**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Simplify $6m + 2n - m + 3n$

SILVER Simplify $4x^2 - 3x + 2x^2 + 7x$

GOLD Simplify $3a^2b + 2ab^2 - a^2b - 5ab^2$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $6m + 2n - m + 3n =$ **$5m + 5n$**

SILVER $4x^2 - 3x + 2x^2 + 7x =$ **$6x^2 + 4x$**

GOLD $3a^2b + 2ab^2 - a^2b - 5ab^2 =$ **$2a^2b - 3ab^2$**

Exam Practice — Collecting Like Terms

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(2 marks)

(a) Simplify $6m + 3n - 2m + n$ (1)

.....

(b) Simplify $5y - 9y + 2$ (1)

.....

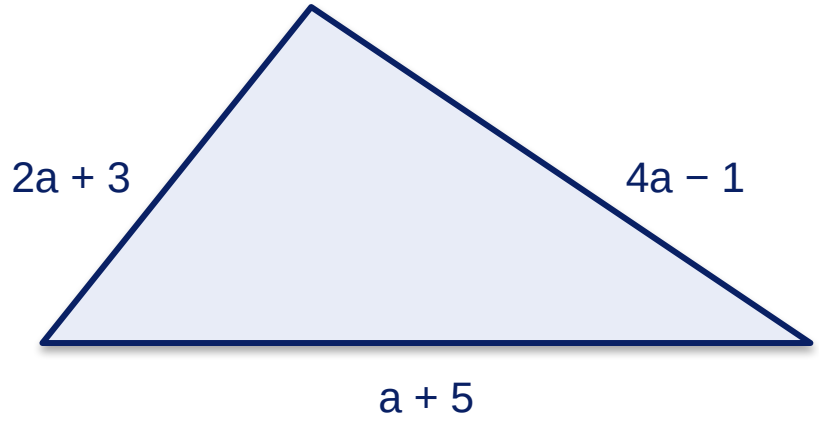
Question 2

(2 marks)

The sides of a triangle are shown.

Find and simplify the perimeter.

Diagram NOT accurately drawn



.....

Exam Practice — Powers and Perimeters

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3

(2 marks)

Simplify

$4x^2 - 3x + 2x^2 + 5x - 7$

.....

Question 4

(2 marks)

The diagram shows a rectangle.

Show that the perimeter is 10p.

Diagram NOT accurately drawn



Exam Practice — Simplifying Fully

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5

(2 marks)

Simplify fully

$3a^2b - 2ab^2 + 5ab - a^2b$

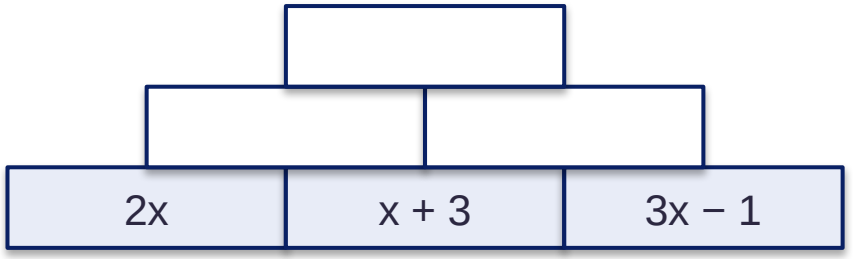
$+ 4ab^2 - 2ab$

.....

Question 6

(3 marks)

In this pyramid, each block is the sum of the two blocks below it.



(a) Find an expression for the top block. (2)

.....

(b) The top block is 47. Find x. (1)

x = ...

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 4m + 4n — B1

(b) −4y + 2 — B1

Tip: the sign belongs to the term after it.

Question 2

2a + 3 + 4a − 1 + a + 5 — M1

= 7a + 7 — A1

Tip: add the letters, then the numbers.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

x^2 terms: $6x^2$, x terms: $2x$ — M1

$6x^2 + 2x - 7$ — A1

Tip: x^2 and x are NOT like terms.

Question 4

$2(3p + q) + 2(2p - q)$ — M1

$6p + 2q + 4p - 2q$

$= 10p$ (shown) — A1

Tip: the q terms cancel out.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

a^2b: $3 - 1 = 2a^2b$ — M1

$2a^2b + 2ab^2 + 3ab$ — A1

Tip: ab^2 and a^2b are different terms.

Question 6

middle: $3x + 3$ and $4x + 2$ — M1

(a) $7x + 5$ — A1

(b) $7x + 5 = 47$, $x = 6$ — B1

Tip: build the pyramid one row at a time.